3- Problema da mochila $O(n \cdot w)$

p	3	2	4	1	$w = 5$
v	9	3	8	6	

i	1	2	3	4	$w = 5$
w_i	3	2	4	1	
v_i	9	3	8	6	

	j					
	0	1	2	3	4	5
0	0	0	0	0	0	0 → ninguém na mochila
1	0	0	0	9	9	9
2	0	0	3	9	9	12
3	0	0	3	9	9	12
4	0	6	6	9	15	(15)

$F(1,3)$ $w_i = 3$, $j = 3$ $w_i \leq j$

$F(i,j) = \max\{F(i-1,j), F(i-1, j-w_i) + v_i\}$

$F(1,3) = \max\{F(0,3), F(0,0) + 9\} = 9$

$F(1,4) = \max\{(0,4), F(0,1)+9\} = 9$

$F(1,5) = \max\{(0,5), F(0,2)+9\} = 9$

$F(2,1) = F(1,1) = 0$

$F(2,2) = \max\{F(1,2), F(1,0)+3\} = 3$

$F(2,3) = \max\{F(1,3), F(1,1)+3\} = 9$

$9 > 0+3$

$F(2,4) = \max\{F(1,4), F(1,2)+3\} = 9$

$F(2,5) = \max\{F(1,5), F(1,3)+3\} = 12$

$9 < (9+3)$

$F(3,1) = w3=4 \quad j=1 \quad w3>j \rightarrow F(i-1,j) = F(2,1) = 0$

$F(3,2) = w3=4 \quad j=2 \quad w3>j \rightarrow F(2,2) = 3$

$F(3,3) = w3=4 \quad j=3 \quad w3>j \rightarrow F(2,3) = 9$

$F(3,4) = w3=4 \quad j=4 ; \max\{F(2,4), F(2,0)+8\} = 9$

$9, \quad 0+8$

$F(3,5) = w3=4 \quad j=5 ; \max\{F(2,5), F(2,1)+8\} = 12$

$12, \quad 0+8$

$F(4,1) = w4=1 \quad j=1 \quad w4<j \rightarrow \max\{F(3,1), F(3,0)+6\} = 6$

$F(4,2) = w4=1 \quad j=2 \quad w4<j ; \max\{F(3,2), F(3,1)+6\} = 6$

$3, \quad 0+6$

$F(4,3) = w4=1 \quad j=3 \quad w4<j ; \max\{F(3,3), F(3,2)+6\} = 9$

$9, \quad 3+6$

$F(4,4) = w4=1 \quad j=4 \quad w4<j ; \max\{F(3,4), F(3,3)+6\} = 15$

$9 \qquad 9+6$

$F(4,5) = w4=1 \quad j=5 \quad w4<j ; \max\{F(3,5), F(3,4)+6\} = 15$

$12 \qquad 9+6$

data
fecha

Alg. guloso

v_i / p_i

$w = 5$ (capacidade)

1	(3)
2	3/2
3	2
4	(6)

Valor ótimo =

item de peso 4 e 1 tem maior fração e juntos somam peso 5. Pela tabela, $v_1 = v_4 = 8$; $v_1 + G = 14$

PD/AG = 15/14 =